

MAJOR PROJECT 2025-2026



B Tech CSE-AIML IV Year II Semester

Department of CSE- Artificial Intelligence and Machine Learning

Major Project Title : KisanSathi: AI-Assisted Farmer Support System

Name of the Internal Guide : Ms. S. Pravallika

Batch Number: 19	
Student Name	Roll Number
Manvitha Pola	22311A6671
Koyyalamudi Rohit Kumar	22311A6670
S. Aashrith Reddy	22311A66C0

INTRODUCTION

- Agriculture is the backbone of India's economy, supporting millions of farmers.
- Farmers often face challenges due to misinformation, middlemen exploitation, and limited access to expert guidance.
- Digital platforms exist, but they are fragmented and not fully accessible to rural communities.
- There is a strong need for a unified system that connects farmers directly with government authorities and experts.
- AI technologies can enhance decision-making through intelligent analysis and automation.
- *KisanSathi* aims to bridge this gap by providing a transparent, AI-assisted digital platform for farmers.

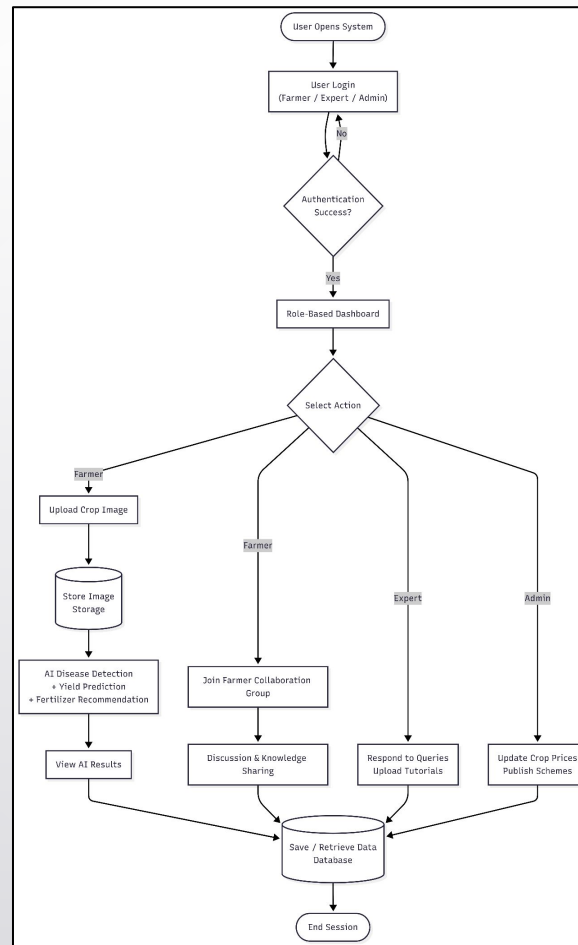
FEATURES

- **Role-Based Access:** Different access for farmers, experts, and government administrators.
- **Verified Crop Prices:** Official crop prices published directly by authorized officials.
- **AI-Powered Insights:** Disease detection, fertilizer recommendations, and yield prediction.
- **Farmer Collaboration:** Crop-based groups for discussion and knowledge sharing.
- **Expert Guidance:** Experts provide tutorials and answer farmer queries.
- **AI Chatbot Support:** Instant assistance for farming-related questions.
- **Multilingual Access:** Content available in local languages for better usability.

SNIST
Greenfield Institute of Science and Technology
Autonomous



DATAFLOW DIAGRAM



METHODOLOGY

- **User Registration & Authentication**

Farmers, experts, and government administrators register and log in through a secure authentication system with role-based access control.

- **Data Collection**

Farmers upload crop images, crop details, and queries. Experts upload tutorials and best practices. Government admins publish verified crop prices and scheme updates.

- **Data Preprocessing**

Uploaded images are cleaned, resized, and normalized before analysis. Text inputs are validated to ensure accuracy and consistency.

- **AI Analysis & Prediction**

AI models perform crop disease detection, crop yield prediction, and fertilizer recommendation based on image and input data.

METHODOLOGY

- **Farmer Collaboration & Expert Support**
Farmers join crop-based groups to discuss challenges and share experiences. Experts respond to queries and validate recommendations.
- **Result Delivery & Advisory Generation**
Prediction results, recommendations, and expert guidance are delivered to farmers through dashboards and chatbot support.
- **Data Storage & Monitoring**
All activities, predictions, messages, and updates are securely stored in the database for tracking, auditing, and system improvement.

MODULE DESIGN

User Management Module

- User registration and login
- Role-based access control (Farmer, Expert, Admin)
- Profile management and authentication

Farmer Module

- Upload crop images and crop details
- View AI predictions and recommendations
- Join collaboration groups and discussions
- Access expert content and chatbot support

Expert Module

- Respond to farmer queries
- Upload tutorials and best practices
- Validate AI recommendations

MODULE DESIGN

Government Admin Module

- Publish verified crop prices
- Update government schemes and announcements
- Monitor platform usage

AI Processing Module

- Crop disease detection
- Crop yield prediction
- Fertilizer recommendation
- Chatbot interaction

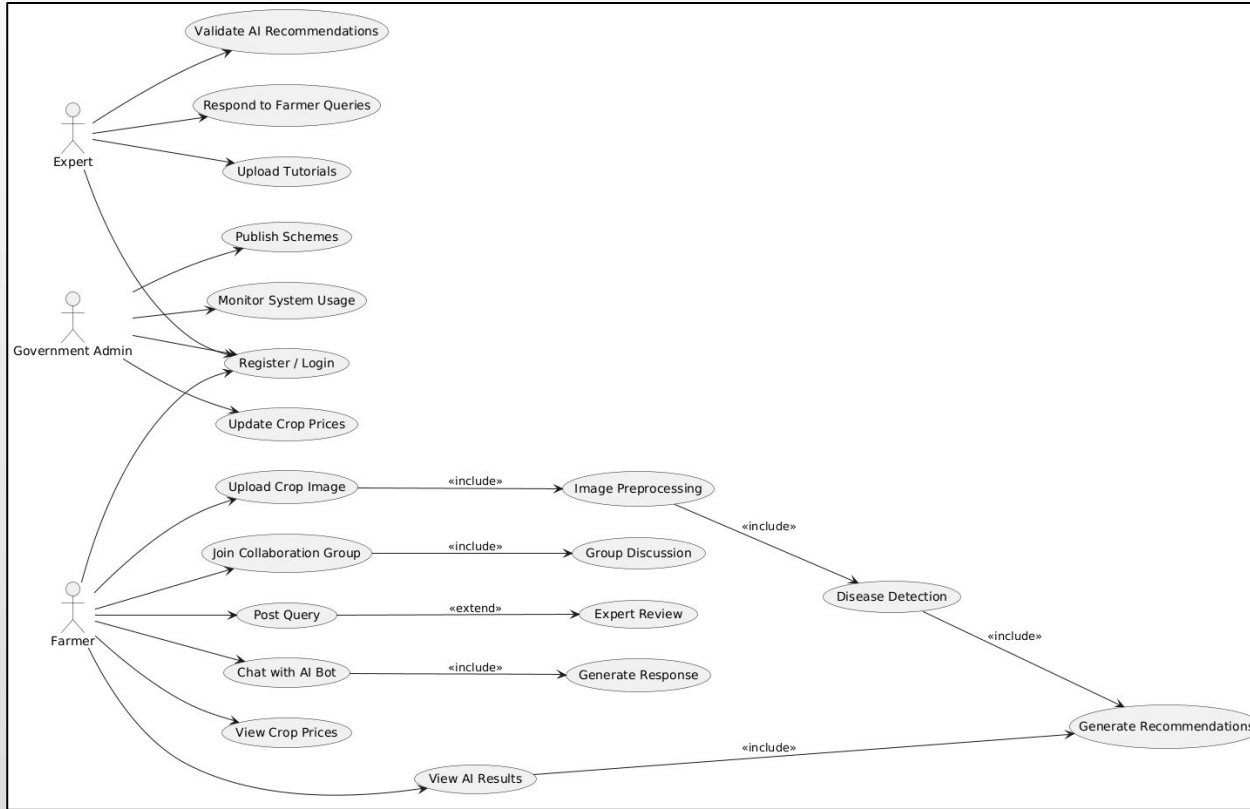
Farmer Collaboration Module

- Crop-based group creation
- Discussion threads and messaging
- Knowledge sharing and moderation

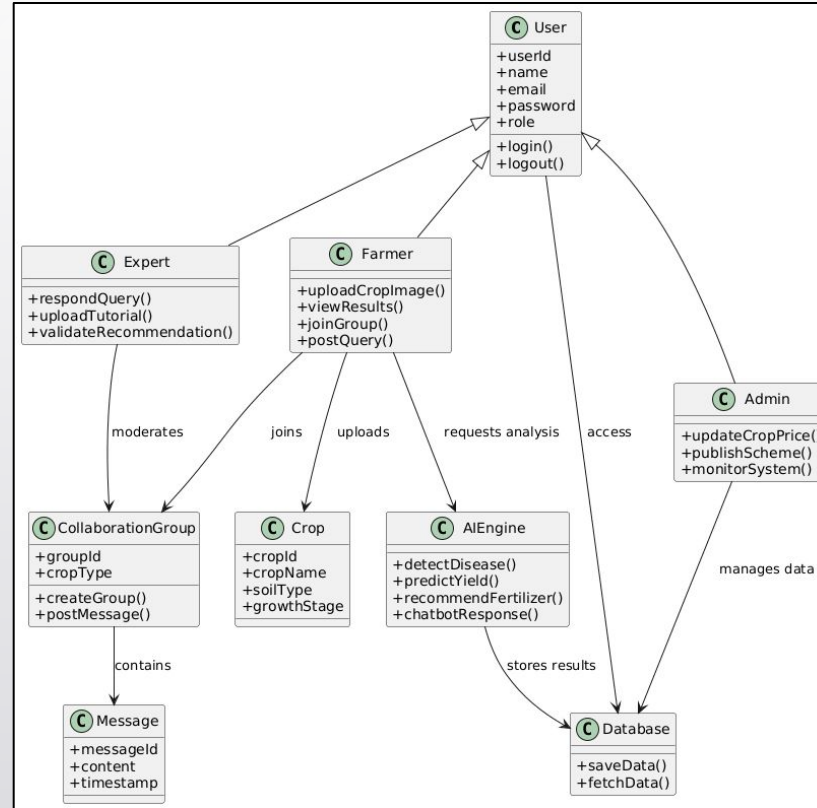
AI INTEGRATION

- **Farmers upload crop images and input details through the web portal.**
- **Requests are sent from the frontend to the Django backend via REST APIs.**
- **Backend forwards data to the AI processing layer for analysis.**
- **Image preprocessing is performed before disease detection.**
- **AI models generate predictions for disease, yield, and fertilizer recommendations.**
- **Chatbot queries are routed to the OpenAI API for intelligent responses.**
- **Results are returned to the backend and displayed on the user dashboard.**
- **All predictions and user interactions are stored in the database.**

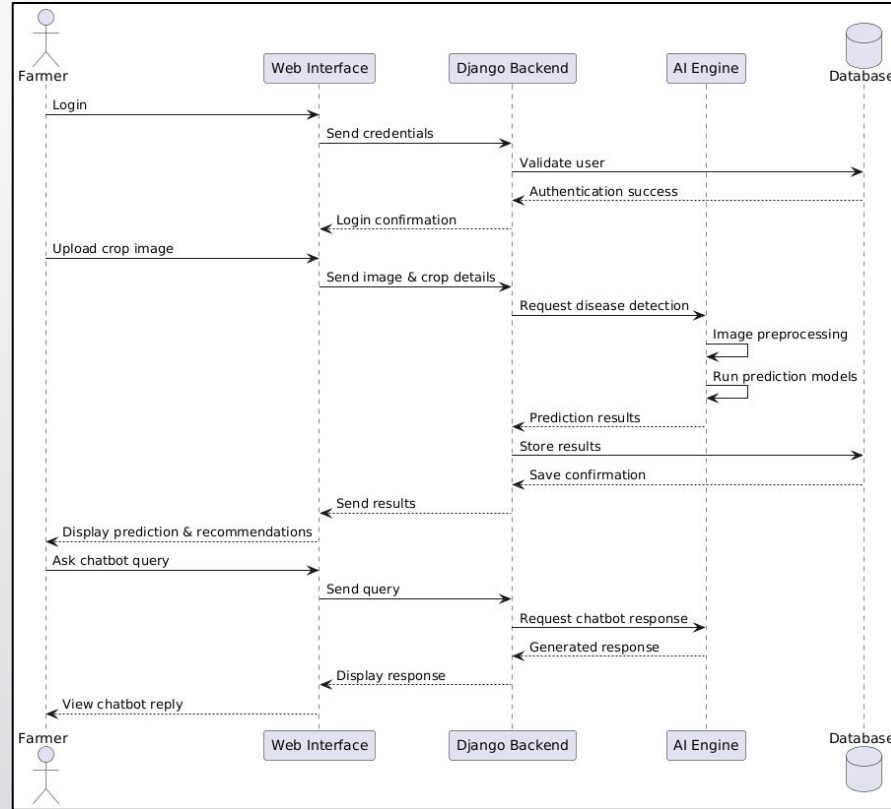
USE CASE DIAGRAM



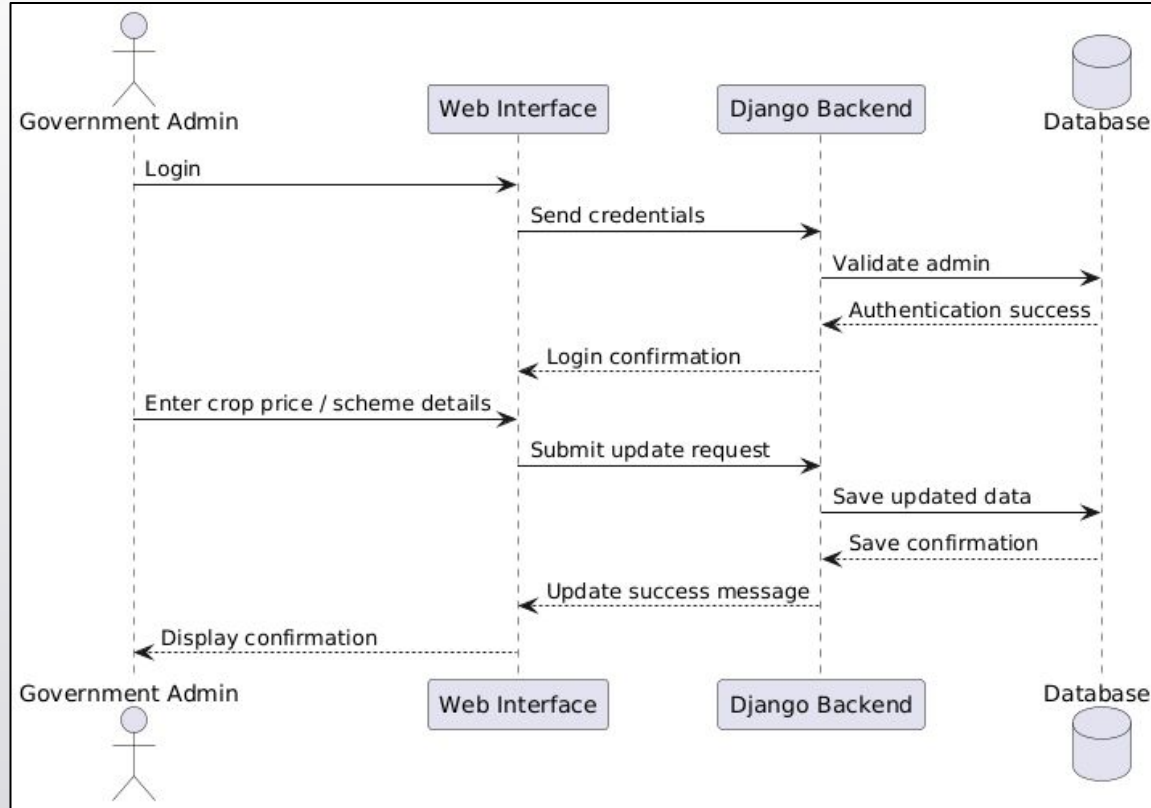
CLASS DIAGRAM



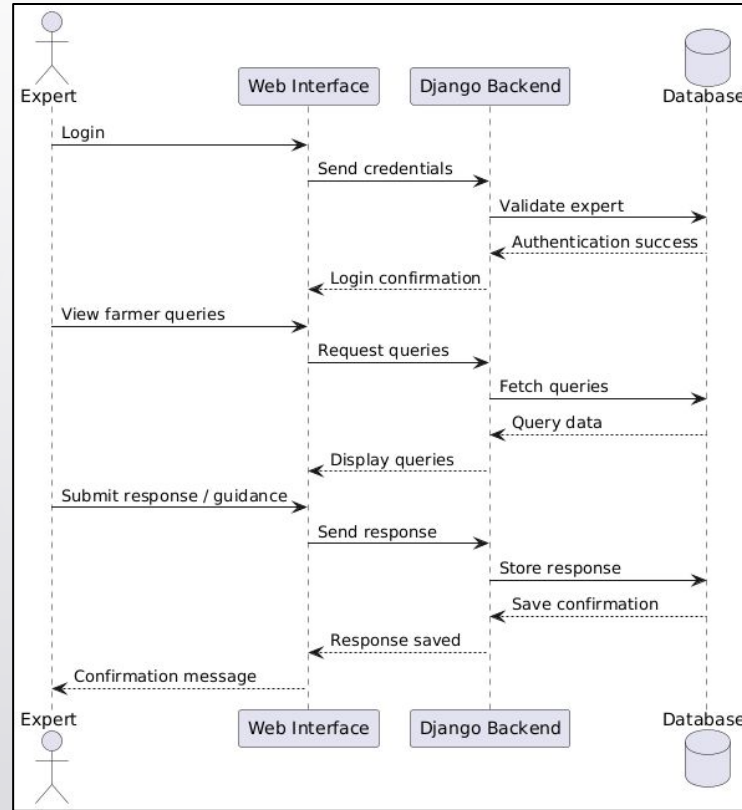
SEQUENCE DIAGRAM (FARMER)



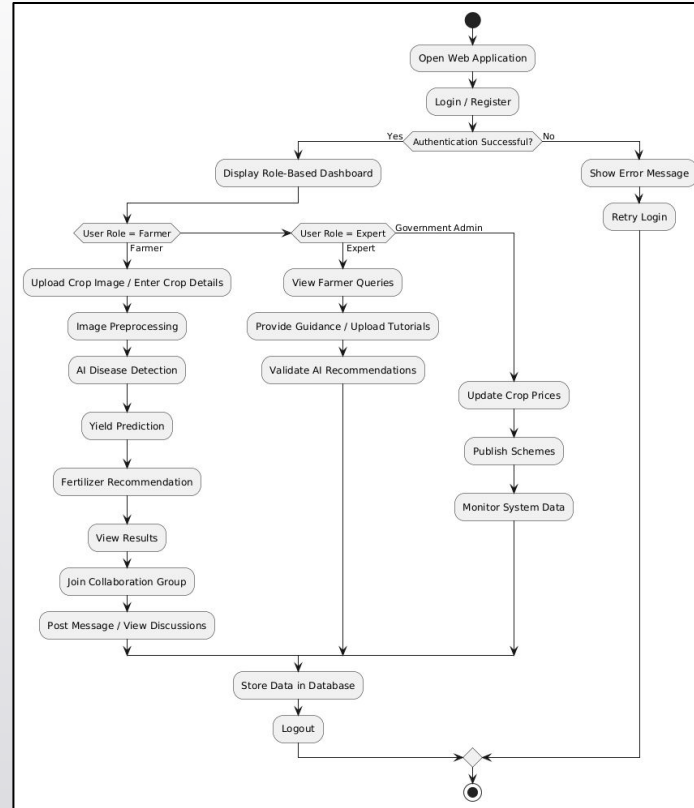
SEQUENCE DIAGRAM (GOVERNMENT ADMIN)



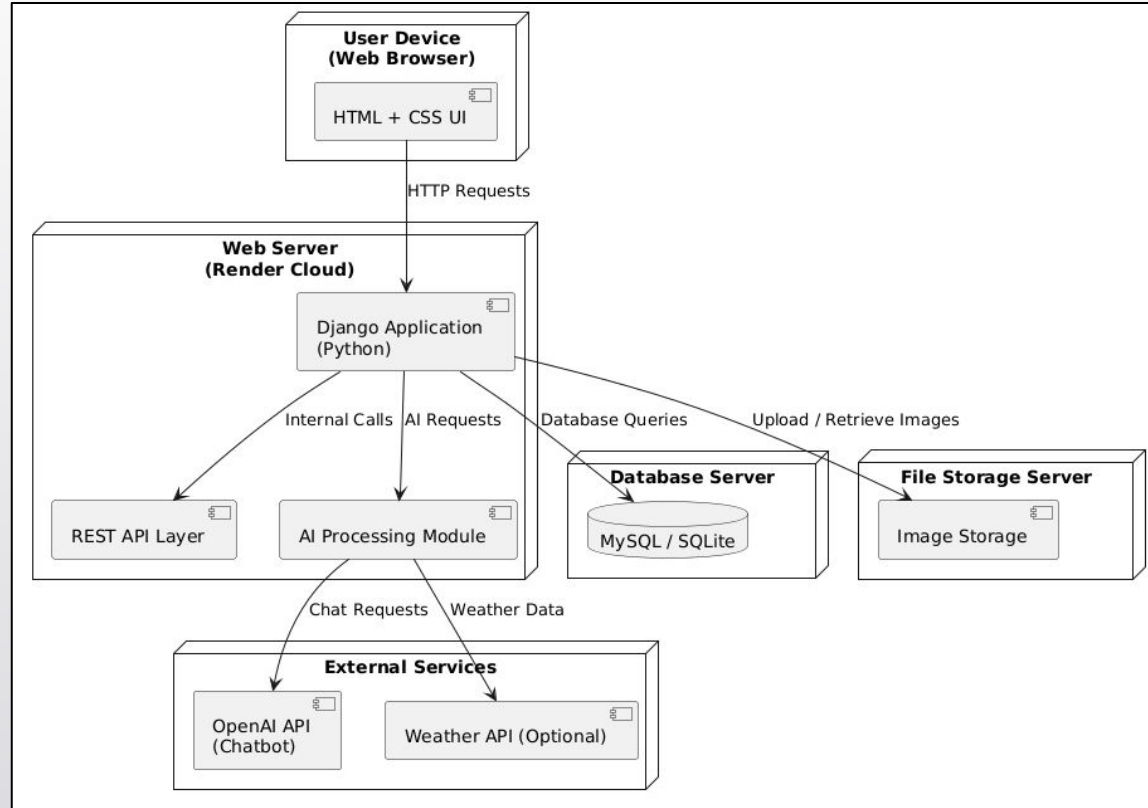
SEQUENCE DIAGRAM (EXPERT)



ACTIVITY DIAGRAM



DEPLOYMENT DIAGRAM



TECHNOLOGY STACK

- **Frontend:** HTML, CSS, Bootstrap
- **Backend:** Python, Django Framework
- **Database:** MySQL
- **AI Frameworks:** TensorFlow, PyTorch, scikit-learn, OpenCV
- **Chatbot Integration:** OpenAI API
- **Hosting & Deployment:** Render
- **Version Control:** GitHub

CONCLUSION

- **KisanSathi provides a unified platform connecting farmers with government authorities and agricultural experts.**
- **The system reduces dependency on middlemen by offering verified and transparent information.**
- **AI-based tools support informed decision-making through disease detection, yield prediction, and recommendations.**
- **Farmer collaboration groups encourage knowledge sharing and community learning.**
- **The platform is accessible, scalable, and suitable for diverse farming communities.**
- **Overall, KisanSathi promotes sustainable agriculture and digital transformation.**

THANK YOU